



THE EVENT AUTOMATION SYSTEM IN THE WIRELESS LAN NETWORK

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**BACHELOR OF ENGINEERING
IN
COMPUTER ENGINEERING**

SCHOOL OF INFORMATION TECHNOLOGY

MAE FAH LUANG UNIVERSITY

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THE EVENT AUTOMATION SYSTEM IN THE WIRELESS LAN NETWORK

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**A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

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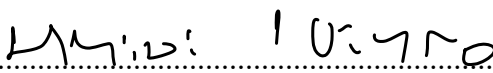
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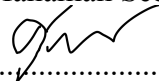
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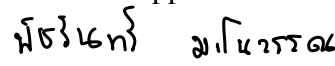
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2022

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Abstract

The objective of our project is to develop a system that facilitates the creation of SSIDs in a convenient and time-saving manner for the personnel within the university. This will be achieved through a user-friendly web application that allows users to generate SSIDs. Users will also be able to select the specific area or location where the SSID will be used, based on various events.

Keyword: wireless access point, authentication, SSID

CONTENTS

SCHOOL OF INFORMATION TECHNOLOGY	1
ACKNOWLEDGEMENT	iii
Abstract	iv
LIST OF TABLES	v
LIST OF FIGURES	vii
CHAPTER 1 INTRODUCTION	1
1.1 Background and Rational	1
1.2 Objective	1
1.3 Scope	1
1.4 Methodology	2
1.5 Plan	2
1.6 Expected Result	2
1.7 Place and Equipment	2
CHAPTER 2 LITERATURE REVIEW	4
2.1 Related Theory	4
2.2 Related Technology	4
CHAPTER 3 ANALYSIS AND DESIGN	7
3.1 Analysis	7
3.2 Activity diagram	8
CHAPTER 4 CONCLUSION	9
4.1 Conclusion	9
REFERENCES	10

LIST OF TABLES

Table	Page
Table 1.1 Working plan	2

LIST OF FIGURES

Figure	Page
Figure 3.2 Activity diagram	8

CHAPTER 1

INTRODUCTION

1.1 Background and Rational

The current process of granting Wi-Fi access to external individuals during university events can be complex and time-consuming. Event organizers need to submit access requests through the university's Information Technology Services Center and manually create detailed SSIDs and passwords for each event. This process needs improvement to become more user-friendly. Therefore, our senior-level project proposes an idea for developing a new Wi-Fi access system. This system will feature a web interface that allows event organizers to register easily, helping in automatically generating SSIDs and passwords in the Wireless LAN Controller (WLC) for each event.

1.2 Objective

- 1.2.1 To develop and study the usage of WLAN with wireless access point.
- 1.2.2 Develop setting and access to wireless access point easier.

1.3 Scope

- 1.3.1 Insider around university.

1.4 Methodology

- 1.4.1 Proposal preparation and defense
- 1.4.2 Literature review and requirement gathering
- 1.4.3 System analysis
- 1.4.4 System design
- 1.4.5 Senior project 1 document preparation
- 1.4.6 Senior Project 1 exam
- 1.4.7 System development
- 1.4.8 System testing
- 1.4.9 Senior Project 2 exam
- 1.4.10 Senior Project 2 exam

1.5 Plan

Table 1.1 Working plan.

Task	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Preproduction									
Production									
Postproduction									

1.6 Expected Result

- 1.6.1 Insider can use our website to enable, create automatic ID and Password for guests and setting range of wireless access point.
- 1.6.2 The system could reduce the workload for IT staff and ensure that each user has a unique and secure login.

1.7 Place and Equipment

1.7.1 Place

- Access point of some building in MFU

1.7.2 Hardware

- Laptop
- ACCESS POINT CISCO CATALYST 9120AX SERIES (C9120AXI-S)

- Cisco Catalyst 9800-40 Wireless Controller

1.7.3 Software

- Microsoft Visual Studio Code
- Python
- Software Defined Network (SDN) - VMware

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

2.1.1 Network security

This theory involves the design, implementation, and maintenance of secure computer networks. The proposed Wi-Fi access system must be designed with network security principles in mind to ensure that only authorized users can access the network, and the system is resistant to cyber-attacks and data breaches.

2.1.2 Authentication and access control

This theory involves verifying the identity of users and granting them access to specific resources based on their privileges. The proposed system must authenticate event organizers and grant them access to the university network based on their permissions and access control policies.

2.1.3 Active Directory

Active Directory (AD) is a directory service protocol used to manage user authentication and authorization. The proposed system will leverage AD server technology to provide a centralized authentication mechanism that simplifies the management of user credentials.

2.1.4 User experience

This theory involves designing systems and interfaces that are intuitive, easy to use, and meet the needs of users. The proposed system must be designed with user experience principles in mind to provide a user-friendly web interface that simplifies the process of requesting.

2.1.5 Web development

This theory involves the development of web applications and the use of programming languages, frameworks, and libraries to build web-based systems. The proposed system will involve the use of web development technologies to build a web interface that enables event organizers to request Wi-Fi access.

2.2 Related Technology

2.2.1 Active Directory

Active Directory is a Microsoft technology that provides centralized authentication and authorization services for Windows-based networks. The proposed system integrates with Active Directory to check and authenticate users, ensuring that only authorized users can access the network.

2.2.2 The 9800-40 WLC

9800-40 WLC supports the latest wireless standards, including 802.11ax (Wi-Fi 6) and 802.11ac Wave 2, providing faster data rates, higher capacity, and improved performance compared to previous generations of wireless technology. It also supports multi-user MIMO and OFDMA technologies, which enable it to serve multiple clients simultaneously and improve overall network efficiency.

The WLC includes advanced features and capabilities to support high-density environments. It can dynamically adjust the channel width, modulation, and coding scheme to optimize performance and capacity based on the number of connected clients and the wireless environment. It also supports band steering and client load balancing, which can help distribute clients across multiple access points to ensure optimal performance.

The 9800-40 WLC provides advanced security features to protect against unauthorized access and attacks. It supports the latest WPA3 security standard and includes features such as MAC address filtering, rogue access point detection, and guest access controls. It also supports Cisco's Identity Services Engine (ISE), which provides policy-based access control and network segmentation based on user and device identity.

2.2.3 Web interface design with RESTful API

A RESTful API is an architectural style for building web services that use HTTP as their underlying communication protocol. REST (Representational State Transfer) is a set of design principles that aim to create simple, lightweight, and scalable web services that can be accessed using standard HTTP methods, such as GET, POST, PUT, and DELETE.

In a RESTful API, each resource is identified by a unique URL, and the HTTP methods are used to perform operations on those resources. For example, a GET request to a specific URL would retrieve a representation of that resource, while a PUT request would update it.

The proposed system's web interface design is based on the principles of user experience and user interface design. It utilizes modern web design technologies and frameworks to create an intuitive and user-friendly interface for event organizers to request Wi-Fi access and generate SSID and passwords in WLC.

2.2.4 Network automation

The proposed system's automated generation of SSID and passwords is related to the concept of automation in network management. Automation is the use of technology to perform tasks and processes automatically, reducing the need for manual intervention and increasing efficiency. The proposed system utilizes network automation to simplify the task of event organizers and reduce the likelihood of errors or inconsistencies.

2.2.5 The C9120AXIS

This supports the latest wireless standards, including 802.11ax (Wi-Fi 6) and 802.11ac Wave 2, which would provide faster data rates and higher capacity compared to previous generations of wireless technology. Its advanced features, such

as multi-user MIMO, OFDMA, and dynamic channel width adjustment, would enable it to serve multiple clients simultaneously and improve overall network efficiency.

In addition, the C9120AXIS includes advanced security features, such as WPA3, MAC address filtering, and rogue access point detection, which would help to protect against unauthorized access and attacks.

Overall, the C9120AXIS access point could be a suitable choice for your proposed Wi-Fi access system project, as its advanced features and capabilities would enable it to provide high-capacity, reliable, and secure wireless connectivity during events at a university.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Analysis

The proposed project aims to streamline the process of granting Wi-Fi access to outsiders during events at the university. Currently, the process is cumbersome and time-consuming, requiring event organizers to manually request access through the Center for Information Technology Services (CITS) and generating SSID and passwords event by event. To address this issue, the proposed project suggests developing a new Wi-Fi access system with a web interface that allows event organizers to easily register and authenticate through an AD server, which automatically generates SSID and passwords in the Wireless LAN Controller (WLC).

The proposed solution leverages several technologies such as the Cisco WLC, RESTful APIs, and AD to provide a secure and reliable method of user authentication, enabling authorized users to access the university network without compromising its security and integrity. By automating the process of generating SSID and passwords in WLC, the proposed system will eliminate the need for manual intervention and simplify the process for event organizers.

One of the strengths of the proposed solution is its ability to streamline the Wi-Fi access process for event organizers, reducing the burden on CITS staff and improving the user experience for event attendees. Additionally, the use of an LDAP server for authentication and authorization provides a secure and reliable method for controlling access to the network.

However, there are also some potential challenges to consider. The development of a new Wi-Fi access system requires a significant investment of time and resources, including the need for skilled developers with expertise in Cisco technologies and AD. Furthermore, there may be compatibility issues between the proposed system and existing network infrastructure, which would need to be addressed before deployment.

In conclusion, the proposed Wi-Fi access system using Cisco WLC, RESTful APIs, and AD has the potential to greatly simplify the process of granting Wi-Fi access during events at the university while improving the user experience and maintaining network security. However, careful planning, development, and testing are necessary to ensure successful implementation and integration with existing network infrastructure.

3.2 Activity diagram

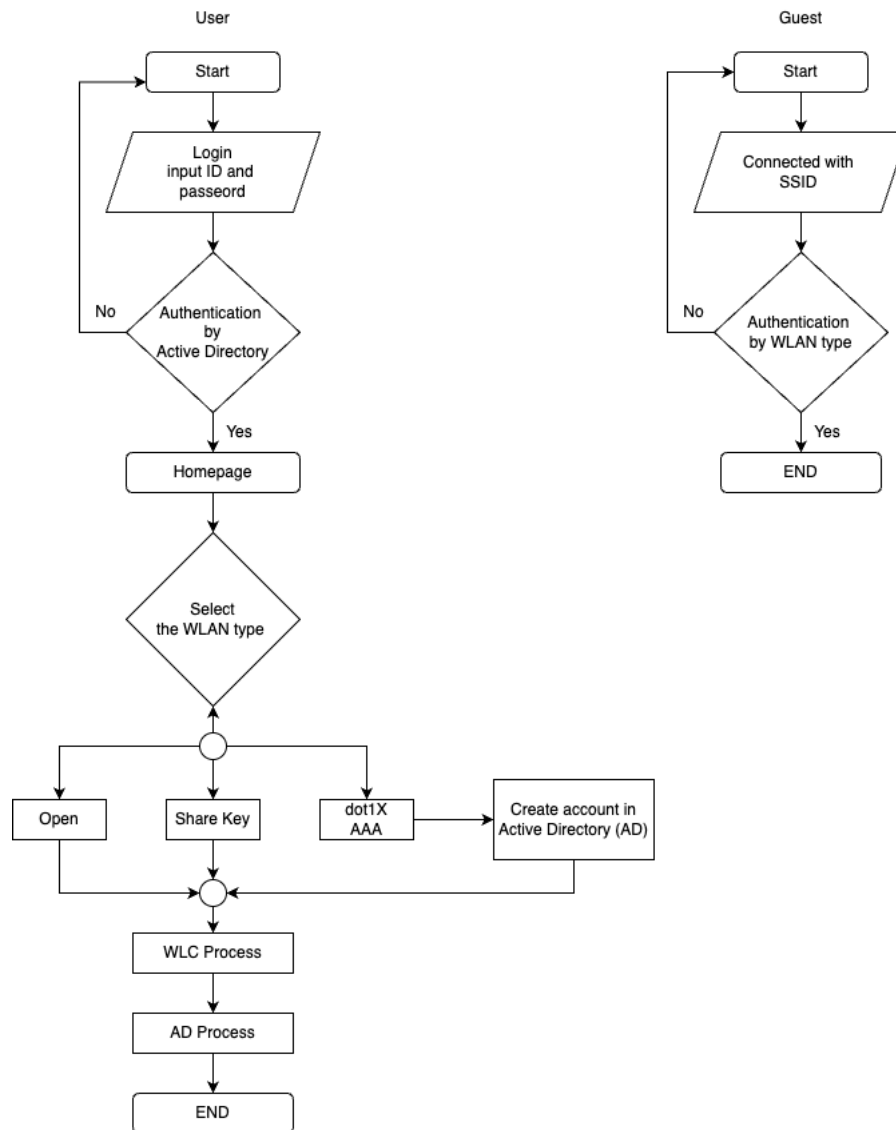


Figure 3.2 Activity diagram

The operation of the event automation system.

CHAPTER 4

CONCLUSION

4.1 Conclusion

Our project is an innovative solution to the issue of creating new SSIDs for staff members at Mae Fah Luang University. By developing a web application that allows staff members to create new SSIDs themselves, we have simplified and streamlined the process, aligning it with the university's goal of transitioning to an e-office system. Our system is user-friendly, scalable, and will save time and resources for both staff members and the Center for Information Technology Services (CITS).

REFERENCES

- [1] Kim, Y., Han, S., & Lee, S. (2018). Dynamic SSID generation for secure wireless communication. *International Journal of Distributed Sensor Networks*, 14(6), 1550147718783706.
- [2] Li, K., Li, Y., Li, J., & Zhao, L. (2019). SSID-based device-free passive indoor localization using deep learning. *IEEE Transactions on Mobile Computing*, 18(4), 774-785.
- [3] Chen, Y., Wang, Q., & Chen, Y. (2020). An intelligent approach to SSID generation for wireless networks. *IEEE Access*, 8, 223722-223731.